

The University of Texas at Austin
IN-CLASS WORK 8

Introduction to Financial Mathematics

September 25, 2026

HEDGING USING EUROPEAN PUT OPTIONS.

Problem 8.1. Farmer Shaun is producing sweet potatoes. He intends to harvest 10,000 —cartons' worth in six months. His total costs are \$12.00 per carton.

He wishes to hedge using European put options. There are two puts on sweet potatoes with the exercise date in six months available: one with the strike price of \$13 per carton and another with the strike price of \$15 per carton. Their premiums are \$0.15 and \$0.18, respectively.

Assume that the prevailing risk-free interest rate is 4% effective for the half-year period.

At harvest time, in six months, it turns out that the sweet-potato spot price equals \$14. What would Farmer Shaun's profit be if he had decided to hedge using the \$13-strike put versus his profit if he had decided to use the \$15-strike put to hedge?

Problem 8.2. Aunt Dahlia simultaneously purchased

- one share of a market index at the current spot price of \$1,000;
 - one one-year, \$1,050-strike put option on the above market index for the premium of \$20.
- i. Is the above portfolio's payoff bounded from above? If you believe it is not, substantiate your claim. If you believe that it is, provide the upper bound.
 - ii. Is the above portfolio's payoff bounded from below? If you believe it is not, substantiate your claim. If you believe that it is, provide the lower bound.

COVERED PUTS.

Problem 8.3. Trader Mavis shorts 10,000 shares of XYZ stock at the current price of \$50.00 per share, to be returned in six months.

She wishes to enhance her bearish position using a covered put. There are two European put options on XYZ stock with the exercise date in six months available: one with the strike price of \$48 per share and another with the strike price of \$45 per share. Their premiums are \$1.20 and \$0.80, respectively.

Assume that the prevailing risk-free interest rate is 2% effective for the half-year period.

In six months, it turns out that XYZ's spot price equals \$46. What would Trader Mavis's profit be if she had decided to write the \$48-strike put versus her profit if she had decided to write the \$45-strike put, in each case to cover her short position?

Problem 8.4. Uncle Baxter simultaneously

- sold short one share of a market index at the current spot price of \$1,000, to be returned in one year;
 - wrote one one-year, \$900-strike put option on the same market index, receiving a premium of \$30.
- i. Is the above portfolio's payoff bounded from above? If you believe it is not, substantiate your claim. If you believe that it is, provide the upper bound.
 - ii. Is the above portfolio's payoff bounded from below? If you believe it is not, substantiate your claim. If you believe that it is, provide the lower bound.

HEDGING USING A LONG PUT AND A SHORT CALL.

Problem 8.5. Grower Percival is producing almonds. He intends to harvest 8,000 pounds' worth in six months. His total costs are \$2.50 per pound.

He wishes to hedge his position using one long options position and one short options position on almonds, both with exercise date in six months. He considers two strategies:

1. Buy a put with strike price \$3.00 per pound (premium \$0.20) and sell a call with strike price \$3.50 per pound (premium \$0.10).
2. Buy a put with strike price \$2.80 per pound (premium \$0.10) and sell a call with strike price \$3.20 per pound (premium \$0.12).

Assume that the prevailing risk-free interest rate is 2.5% effective for the half-year period.

At harvest time, in six months, it turns out that the almond spot price equals \$3.55. What would Grower Percival's profit be under the first strategy versus his profit under the second strategy?

Problem 8.6. Cousin Reginald simultaneously purchased

- one share of a market index at the current spot price of \$1,000;
 - one one-year, \$900-strike put option on the above market index for the premium of \$15;
 - and also wrote one one-year, \$1,100-strike call option on the same market index, receiving a premium of \$20.
- i. Is the above portfolio's payoff bounded from above? If you believe it is not, substantiate your claim. If you believe that it is, provide the upper bound.
 - ii. Is the above portfolio's payoff bounded from below? If you believe it is not, substantiate your claim. If you believe that it is, provide the lower bound.

HEDGING USING A SHORT PUT AND A LONG CALL.

Problem 8.7. Confectioner Wilhelmina is producing chocolate and will need to purchase 6,000 pounds of cocoa in six months to fulfill her production needs.

She wishes to hedge against a rise in the price of cocoa using a long call together with a short put, both with exercise date in six months. She considers two strategies:

1. Buy a call with strike price \$4.00 per pound (premium \$0.25) and sell a put with strike price \$3.50 per pound (premium \$0.15).
2. Buy a call with strike price \$4.20 per pound (premium \$0.15) and sell a put with strike price \$3.70 per pound (premium \$0.20).

Assume that the prevailing risk-free interest rate is 3% effective for the half-year period.

In six months, it turns out that the cocoa spot price equals \$4.50. What would Confectioner Wilhelmina's effective cost of acquiring the cocoa be under the first strategy versus the second strategy?

